

P1

Measurement

Physics | Classes 6, 7 & 8 Combined

Classes 6 • 7 • 8 | TET Paper II Science

A • MEASUREMENT — Definition & Need

Measurement	Comparison of an unknown quantity with a known standard quantity — result has TWO parts: a number and a unit
Need for Measurement	For uniformity and accuracy across countries, scientists adopted a common system (SI System)
Physical Quantity	Any quantity that can be measured — e.g., length, mass, time, temperature, current
Fundamental Quantity	Independent physical quantity that cannot be derived from others — 7 in SI system
Derived Quantity	Quantity derived from fundamental quantities — e.g., area = length × length; speed = length/time
Unit	Known standard used for comparison in measurement — e.g., metre, kilogram, second

B • UNIT SYSTEMS — FPS, CGS, MKS & SI

Syst em	Full Form	Length	Mass	Time	Type / Notes
FPS	Foot-Pound-Second	Foot (ft)	Pound (lb)	Second (s)	British system — NOT a metric system
CGS	Centimetre-Gram-Second	Centimetre (cm)	Gram (g)	Second (s)	Metric system
MKS	Metre-Kilogram-Second	Metre (m)	Kilogram (kg)	Second (s)	Metric system — precursor to SI
SI	Système International (International System)	Metre (m)	Kilogram (kg)	Second (s)	Metric — universally accepted; adopted in 1960 at Paris

■ FPS = British system (NOT metric). CGS, MKS, SI = metric systems. TET exams test this difference!

■ Did You Know?

- ◆ In December 1998, NASA's Mars Climate Orbiter (worth \$125 million) was lost because one team used FPS units and another used MKS units — unit mismatch caused the spacecraft to approach Mars at wrong altitude and crash.
- ◆ SI stands for "Système International" — French words. The system was established at the 11th General Conference on Weights and Measures in Paris, France in 1960.
- ◆ The metric system was created by the French in 1790.

C · SI BASE QUANTITIES — All 7 Fundamental Units

Quantity	SI Unit	Symbol	Measuring Instrument	Class
Length	metre	m	Ruler / Scale / Metre rod / Vernier calliper / Screw gauge	6,7,8
Mass	kilogram	kg	Beam balance / Electronic balance / Spring balance	6,7,8
Time	second	s	Clock / Stopwatch / Pendulum	6,7,8
Temperature	kelvin	K	Thermometer (clinical, lab, digital)	8
Electric Current	ampere	A	Ammeter	8
Amount of Substance	mole	mol	Calculated from mass and molar mass	8
Luminous Intensity	candela	cd	Photometer (Luminous Intensity Meter)	8

Supplementary Quantities — Plane Angle & Solid Angle

Quantity	Definition	SI Unit	Symbol	Dimension
Plane Angle	Angle at intersection of two straight lines or planes	radian	rad	2D — two dimensional
Solid Angle	Angle at intersection of three or more planes at a common point; also = angle at vertex of a cone	steradian	sr	3D — three dimensional

- π radian = 180° | 1 radian = $180^\circ/\pi$ | $1^\circ = \pi/180$ radian
- Radian: angle at centre of circle where arc length = radius of circle
- Steradian: solid angle at centre of sphere where surface area = (radius)²
- Until 1995, plane angle and solid angle were "supplementary quantities" — reclassified as derived quantities in 1995

D · LENGTH — Units, Conversions & Measurement

Length Conversions — Memorise All

Conversion	Value	Direction
1 km → m	1,000 m (10^3 m)	km to m: $\times 1000$
1 m → cm	100 cm (10^2 cm)	m to cm: $\times 100$
1 cm → mm	10 mm	cm to mm: $\times 10$
1 m → mm	1,000 mm	m to mm: $\times 1000$
1 km → cm	1,00,000 cm (10^5 cm)	km to cm: $\times 100,000$
1 km → mm	10,00,000 mm (10^6 mm)	km to mm: $\times 1,000,000$
1 nautical mile	= 1.852 km	Sea/air navigation

SI Prefixes — Multiples & Submultiples

Prefix	Symbol	Meaning	Value	For Metre
Nano	n	Submultiple: $1/10^9$	10^{-9}	1,000,000,000 nm = 1 m
Micro	μ	Submultiple: $1/10^6$	10^{-6}	10,00,000 μ m = 1 m
Milli	m	Submultiple: $1/1,000$	10^{-3}	1,000 mm = 1 m
Centi	c	Submultiple: $1/100$	10^{-2}	100 cm = 1 m
Deci	d	Submultiple: $1/10$	10^{-1}	10 dm = 1 m
Kilo	k	Multiple: 1,000	10^3	1 km = 1,000 m
Mega	M	Multiple: 10,00,000	10^6	1 Mm = 10^6 m

Measuring Instruments for Length

Metre Scale / Ruler	Measures straight lengths — range: 0 to 15 cm (small) or 0 to 30 cm (large) or 1 metre
Measuring Tape	Flexible tape for curved body measurements (chest, waist) — used by tailors; not for straight lines
Divider Method	Measures curved lines by stepping equal segments along the curve then counting segments $\times$ length
String Method	Place string along curved line, straighten it, then measure with ruler — finds length of curved line

Vernier Callipers	Precise measurement of small lengths, inner/outer diameters — least count = 0.1 mm
Screw Gauge	Very precise measurement of thin wires, thickness of paper — least count = 0.01 mm
Odometer	Device on automobiles that indicates total distance travelled by the vehicle

Measurement Errors & Corrections

Parallax Error	Error due to incorrect eye position — eye must be VERTICALLY ABOVE the measurement point, not tilted
Zero Error	Error when the instrument does not read zero when nothing is measured — must be corrected before use
Correct method	Head of object at zero of scale Eye vertically above Count cm then mm from the end

- Never hold the thermometer by the bulb; never heat it in flame or sun (for temperature instruments)
- Always keep object PARALLEL to the scale when measuring length
- Standard metre rod: made of platinum-iridium alloy at Bureau of Weights and Measures, Paris, France
- India's copy of standard metre rod is at the National Physical Laboratory (NPL), New Delhi

E · MASS — Definition, Units & Instruments

Mass	Measure of the amount of matter in an object — SI unit: kilogram (kg) — does NOT change with location
Weight	Gravitational pull experienced by matter — $\text{Weight} = \text{mass} \times g$ — changes with location (moon vs earth)
Key Difference	Mass = constant everywhere Weight = varies (1/6 on Moon vs Earth)

Mass Conversions

Conversion	Value
1 kg → g	1,000 g
1 g → mg	1,000 mg
1 tonne → kg	1,000 kg
1 quintal → kg	100 kg

Conversion	Value
Standard 1 kg	Mass of platinum-iridium bar at International Bureau of Weights and Measures, Sèvres, France (since 1889)

Instruments for Mass Measurement

Beam Balance	Compares unknown mass with known standard masses — works by equilibrium principle
Electronic Balance	Precise digital measurement — used in laboratories for chemicals, in shops for groceries and jewellery
Spring Balance	Uses spring extension to measure weight (in Newtons) — not for precise mass measurement

F · TIME — Units, Instruments & Clock Types

Time Units & Conversions

Conversion	Value
1 minute	= 60 seconds
1 hour	= 60 minutes = 3,600 seconds
1 day	= 24 hours = 86,400 seconds
1 year	= 365 days = 8,760 hours
SI Unit	Second (s) — represented as 's'

Types of Clocks (Class 8 — Section 1.3)

By Display Type:

Analog Clock

- Classic clock with three hands
- Hours hand — short and thick
- Minutes hand — long and thin
- Seconds hand — long, very thin; makes 1 rotation per minute; 60 rotations per hour
- Driven by mechanical or electronic mechanism

Digital Clock

- Shows time directly in numerals or symbols
- May display 12-hour or 24-hour format
- Modern digital clocks show date, day, month, year, temperature

By Working Mechanism:

Quartz Clock

- Uses electronic oscillations controlled by a quartz crystal (SiO₂)
- Crystal vibration frequency is very precise
- More accurate than mechanical clocks
- Accuracy: 1 second error per 10⁹ seconds

Atomic Clock

- Uses periodic vibrations occurring within atoms
- Most accurate clock ever made
- Accuracy: 1 second error per 10¹³ seconds
- Used in GPS, GLONASS, International Time Distribution Services

Historical & Special Timekeeping

- Ancient time measurement: sand clock (hourglass) and sundial — used during daytime
- Pulse measurement: counting heartbeats ($\approx 75/\text{min}$) gives approximate time elapsed

GMT	Greenwich Mean Time — mean solar time at Royal Observatory, Greenwich, London (0° longitude)
IST	Indian Standard Time = GMT + 5:30 hours Reference: Mirzapur, Uttar Pradesh (82.5°E longitude)
Time Zones	Earth divided into 24 zones $\times 15^\circ$ longitude each Difference between adjacent zones = 1 hour

G · Additional Base Quantities — Temperature, Current, Mole, Candela

Temperature	Measure of average KE of particles — degree of hotness/coldness — SI unit: kelvin (K) — measured by thermometer
Electric Current (I)	Flow of electric charges in a direction $I = Q/t$ (charge/time) — SI unit: ampere (A) — measured by ammeter
1 Ampere	If 1 coulomb of charge flows through conductor in 1 second $\rightarrow$ current = 1 ampere
Amount of Substance	Measure of number of entities (atoms, molecules, ions) — SI unit: mole (mol)
1 Mole	$= 6.023 \times 10^{23}$ entities — known as Avogadro Number — mole is used because atoms are too small to count individually
Luminous Intensity	Measure of power of emitted light in a particular direction per unit solid angle — SI unit: candela (cd)
1 Candela	$\approx$ light emitted by one common wax candle — measured by photometer (Luminous Intensity Meter)
Luminous Flux	Measure of total perceived power of light — SI unit: lumen (lm)
Calibration	Process of configuring an instrument to measure accurately within a specific range

H · Accuracy, Precision, Approximation & Rounding Off

Concept	Definition	Example / Remember
Error	Difference between the real (true) value and the observed (measured) value	Every measurement has some uncertainty — cannot be completely eliminated

Concept	Definition	Example / Remember
Accuracy	Closeness of a measured value to the actual/true value	Arrows all hitting bullseye = accurate AND precise (best case)
Precision	Closeness of two or more measurements to EACH OTHER (repeatability)	Arrows all hitting same spot (but not centre) = precise but NOT accurate
Approximation	Finding a number acceptably close to exact value by estimation — used when data is inadequate	Heart beats $\approx 75/\text{min}$ $\rightarrow$ in a day $\approx 75 \times 60 \times 24 = 1,08,000$ beats (approximate)
Rounding Off	Technique to reduce digits in a result to manageable number	1.864 $\rightarrow$ 1.86 (next digit < 5 , keep 6) 1.868 $\rightarrow$ 1.87 (next digit > 5 , increase 6 to 7)

Rules for Rounding Off

- Step 1: Decide which is the LAST DIGIT to keep
- Step 2: If the NEXT digit is less than 5 $\rightarrow$ keep last digit SAME
- Step 3: If the NEXT digit is 5 or greater than 5 $\rightarrow$ INCREASE last digit by 1
- Example: Round 4.582 to 2 decimal places $\rightarrow$ last digit = 8, next = 2 (< 5) $\rightarrow$ answer: 4.58
- Example: Round 4.586 to 2 decimal places $\rightarrow$ last digit = 8, next = 6 (> 5) $\rightarrow$ answer: 4.59

■ Accuracy vs Precision — TET Trap

- Accurate = close to TRUE value | Precise = close to EACH OTHER (can be precise but wrong!)
- Good precision + good accuracy = all arrows at bullseye (ideal measurement)
- Good precision + poor accuracy = arrows clustered together but far from centre
- Poor precision + poor accuracy = arrows scattered AND far from centre

I · Solved Numerical Problems

Problem 1	The distance between two cities is 43.65 km. Convert to metres and centimetres.
Given	$d = 43.65 \text{ km}$
Solution	In metres: $43.65 \times 1000 = 43,650 \text{ m}$ In cm: $43,650 \times 100 = 43,65,000 \text{ cm}$
Answer	43,650 m = 43,65,000 cm
Problem 2	Length of pencil: scale reading at one end = 2.0 cm, other end = 12.1 cm. Find length.
Given	Reading 1 = 2.0 cm; Reading 2 = 12.1 cm

Solution	Length = $12.1 - 2.0 = 10.1$ cm
Answer	10.1 cm = 101 mm

Problem 3	Distance from school to home = 2250 m. Express in kilometres.
Given	d = 2250 m
Solution	d in km = $2250 / 1000 = 2.250$ km
Answer	2.25 km

Problem 4	Convert 60° into radians.
Given	$1^\circ = \pi/180$ radian
Solution	$60^\circ = \pi/180 \times 60 = \pi/3$ radian = $3.14/3 \approx 1.047$ rad
Answer	$\pi/3$ radian ≈ 1.047 rad

Problem 5	If 2 coulombs of charge flows through a circuit for 10 seconds, calculate the current.
Given	Q = 2 C; t = 10 s
Solution	I = $Q/t = 2/10 = 0.2$ A
Answer	Current = 0.2 Ampere

Problem 6	Round off 7875 cm to km and m.
Given	7875 cm
Solution	$7875 \text{ cm} \div 100 = 78.75 \text{ m} \rightarrow 78 \text{ m } 75 \text{ cm} \mid 78.75 \text{ m} \div 1000 = 0.07875 \text{ km}$
Answer	78 m 75 cm = 0.07875 km

J · Key Facts — Rapid Revision (Numbers, Years & Names only)

Values, Formulas & Unit Conversions

Quantity / Concept	Formula / Value	Unit / Note
Measurement	Number + Unit (two parts)	e.g., 5 kg; 3 metres
1 km	= 1,000 m = 1,00,000 cm = 10,00,000 mm	Length
1 m	= 100 cm = 1,000 mm	Length
1 kg	= 1,000 g 1,000 kg = 1 tonne	Mass
1 hour	= 3,600 s 1 day = 86,400 s	Time

Quantity / Concept	Formula / Value	Unit / Note
π radian	$= 180^\circ \rightarrow 1^\circ = \pi/180 \text{ rad}$	Plane angle
Avogadro Number	6.023×10^{23} per mole	Amount of substance
1 candela	$\approx$ light of one common wax candle	Luminous intensity
Electric current	$I = Q/t$ (charge/time)	Ampere (A)
1 Ampere	$= 1$ coulomb per second	Definition
Quartz clock accuracy	1 second error per 10^9 seconds	Class 8
Atomic clock accuracy	1 second error per 10^{13} seconds	Class 8 — most accurate
IST	GMT + 5:30 hours Reference: 82.5°E longitude	Mirzapur, UP
Time zone width	15° longitude = 1 hour difference	24 zones for 360°
Body temperature	$37^\circ\text{C} = 98.6^\circ\text{F} = 310.15 \text{ K}$	Normal human
Rounding: next digit < 5	Keep last digit SAME	Rule
Rounding: next digit ≥ 5	Increase last digit by 1	Rule

Scientists, Events & Years

Scientist / Event	Contribution / Fact	Year / Class
SI System established	11th General Conference on Weights & Measures, Paris, France	1960
Metric system created	French scientists established the metric system	1790
Standard metre rod	Platinum-iridium alloy at Bureau of Weights & Measures, Paris; India's copy at NPL, New Delhi	Cl. 6
Standard 1 kg	Platinum-iridium bar at International Bureau of Weights & Measures, Sèvres, France	1889
Ruler inventor	William Bedwell invented the ruler in the 16th century	16th century
Odometer	Device indicating distance travelled by automobile	Cl. 6
Mars Orbiter disaster	NASA lost \$125M Mars Climate Orbiter due to FPS vs MKS unit mismatch between two teams	Sept 23, 1999
GMT reference	Greenwich Mean Time at Royal Observatory, Greenwich, London at 0° longitude	Cl. 8

K · Glossary — Definitions

Measurement — Comparison of unknown quantity with known standard; result has number + unit	Physical Quantity — Any quantity that can be measured — e.g., length, mass, time, temperature
Fundamental Quantity — Independent physical quantity — 7 in SI: length, mass, time, temp, current, amount, luminous intensity	Derived Quantity — Quantity derived from fundamental quantities — e.g., area, volume, density, speed
SI System — International System of Units — "Système International" — adopted in 1960 at Paris	FPS System — British system: Foot, Pound, Second — NOT a metric system
CGS System — Metric system: Centimetre, Gram, Second	MKS System — Metric system: Metre, Kilogram, Second — precursor to SI
Metre (m) — SI unit of length	Kilogram (kg) — SI unit of mass
Second (s) — SI unit of time	Kelvin (K) — SI unit of temperature
Ampere (A) — SI unit of electric current — 1 A = 1 coulomb per second	Mole (mol) — SI unit of amount of substance — 1 mol = 6.023×10^{23} entities (Avogadro Number)
Candela (cd) — SI unit of luminous intensity — $\approx$ light from one wax candle	Radian (rad) — SI unit of plane angle — angle where arc length = radius of circle
Steradian (sr) — SI unit of solid angle — 3D angle at vertex of cone / intersection of 3+ planes	Parallax Error — Error due to eye not being directly above measurement point — eye displacement causes wrong reading
Zero Error — Instrument error when it does not read zero before measurement begins — must be subtracted or added	Accuracy — Closeness of measured value to the true value
Precision — Closeness of two or more measurements to each other — repeatability	Approximation — Process of finding a value acceptably close to exact value by estimation
Rounding Off — Reducing digits in a number: if next digit < 5, keep same; if ≥ 5 , increase by 1	Calibration — Process of configuring an instrument to measure accurately within a specific range
Beam Balance — Instrument to measure mass by comparing unknown mass with known standard masses	Electronic Balance — Precise digital instrument for measuring mass — used in labs and shops
Vernier Callipers — Precise instrument to measure small lengths — least count = 0.1 mm	Screw Gauge — Very precise instrument for measuring thin wires — least count = 0.01 mm
Odometer — Device in automobiles that records total distance travelled	Quartz Clock — Clock using electronic oscillations from quartz crystal — accuracy: 1s error per 10^9 s
Atomic Clock — Most accurate clock using atomic vibrations — accuracy: 1s error per 10^{13} s; used in GPS	GMT — Greenwich Mean Time — mean solar time at Royal Observatory, London, at 0° longitude

IST — Indian Standard Time = GMT + 5:30 hours; reference: Mirzapur (82.5°E), Uttar Pradesh	Avogadro Number — 6.023×10^{23} — number of atoms/molecules in one mole of any substance
Photometer — Instrument to measure luminous intensity in candela	Ammeter — Instrument to measure electric current in ampere — connected in series in a circuit